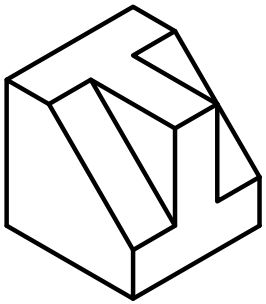
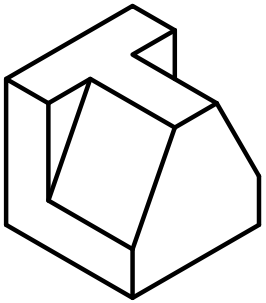
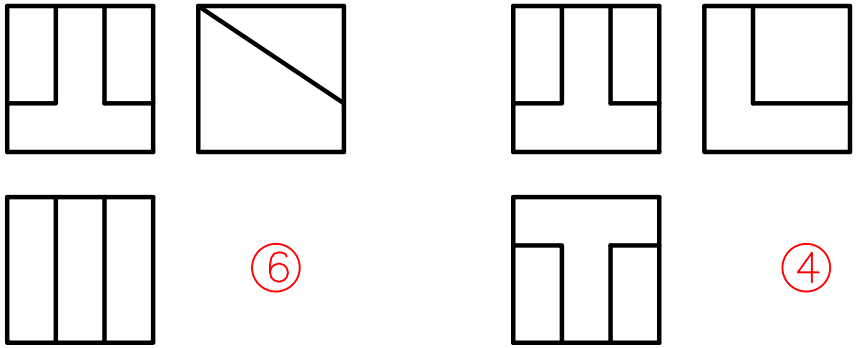
3—1　根据立体图所表示的物体形状,找出对应的三面投影图。

①



②

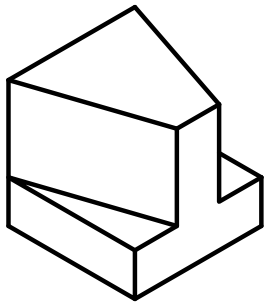




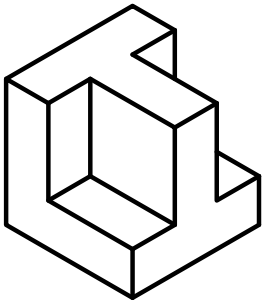
⑥

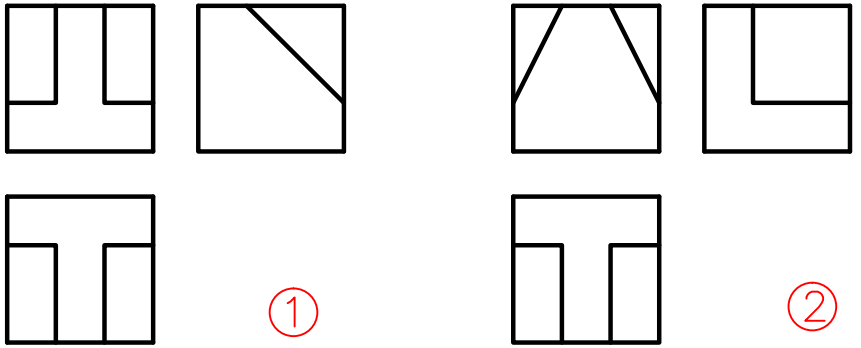
④

③



④

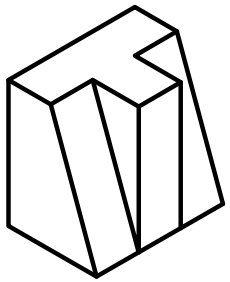




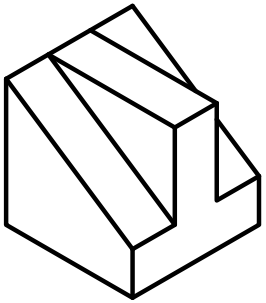
①

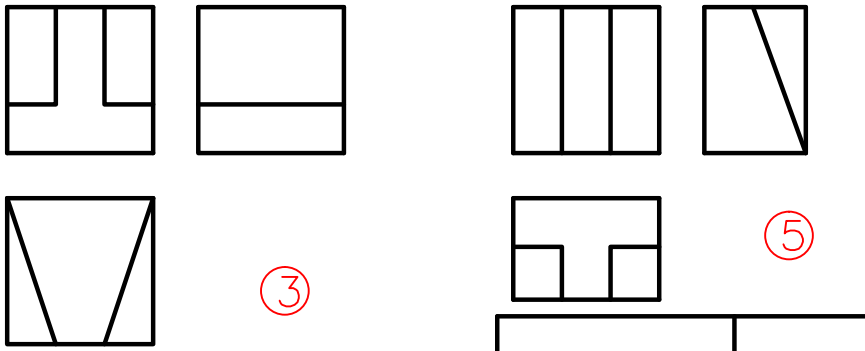
②

⑤



⑥

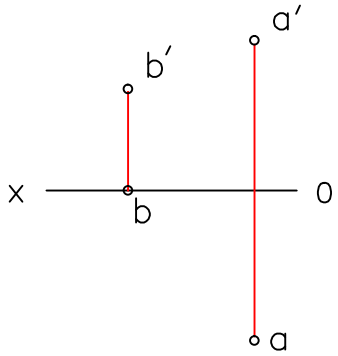


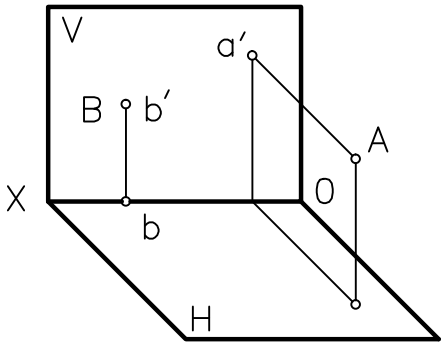


③

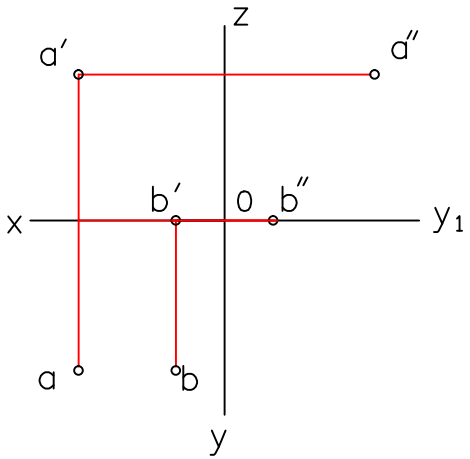
⑤

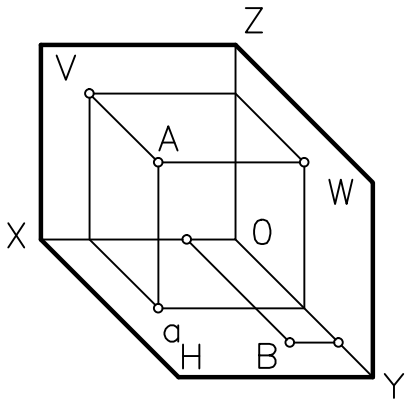
3－2　已知各点的空间位置，求作投影图，并填写出各点到投影面的距离（单位为mm）。





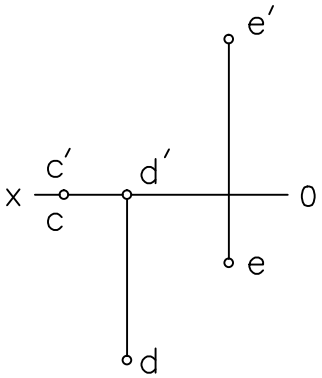
	H	V
A	15	15
B	10	0

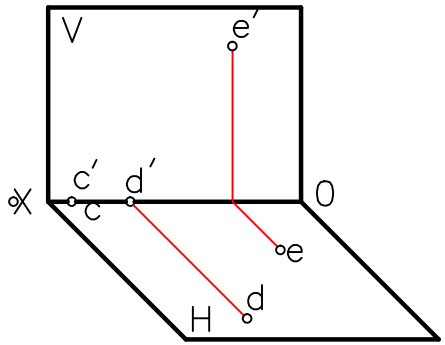




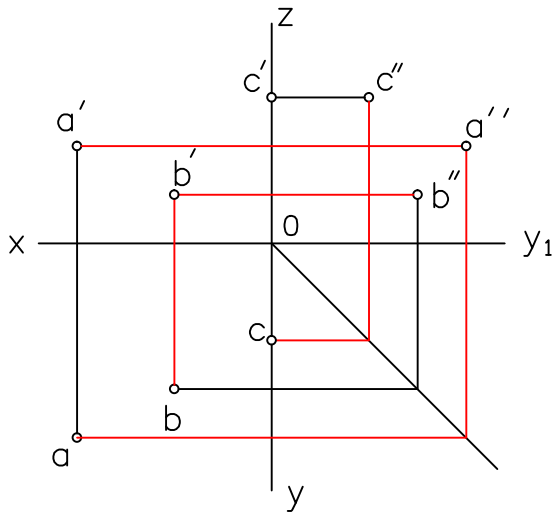
	H	V	W
A	15	10	15
B	0	15	5

3－3　根据投影图，画出各点的空间位置。



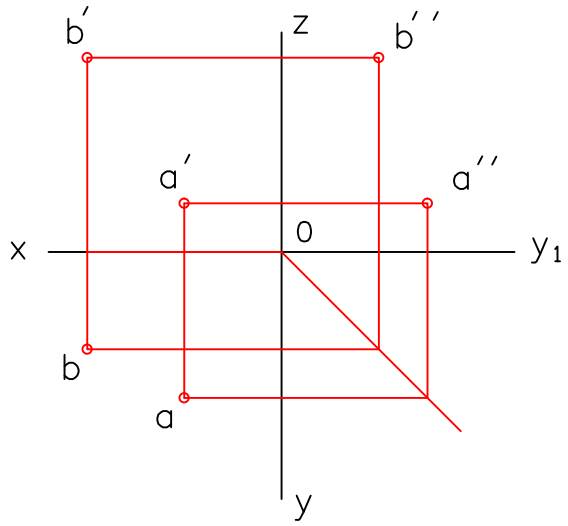


3－4　求各点的第三面投影，并填写各点到投影面的距离。



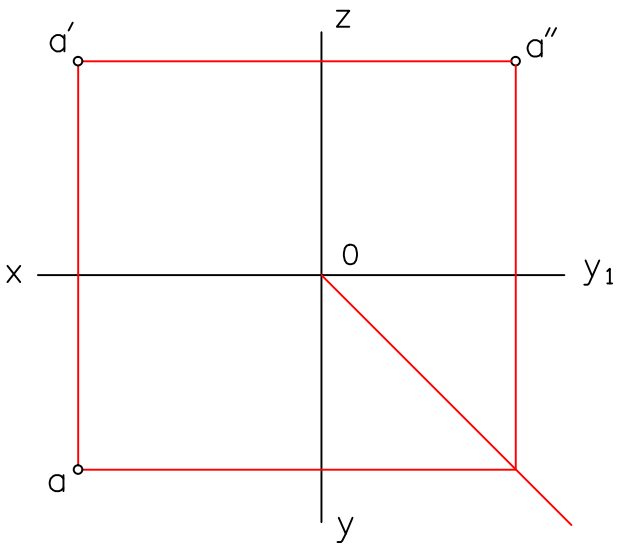
	H	V	W
A	10	20	20
B	5	15	10
C	15	10	0

3－5　已知各点的坐标值，求其三面投影图。

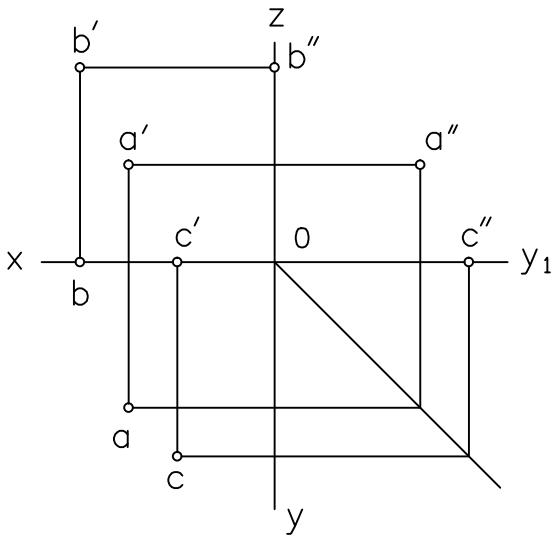


	x	y	z
A	10	15	5
B	20	10	20

3－6　已知A点的侧面投影，及XA=25，求其另两面投影。

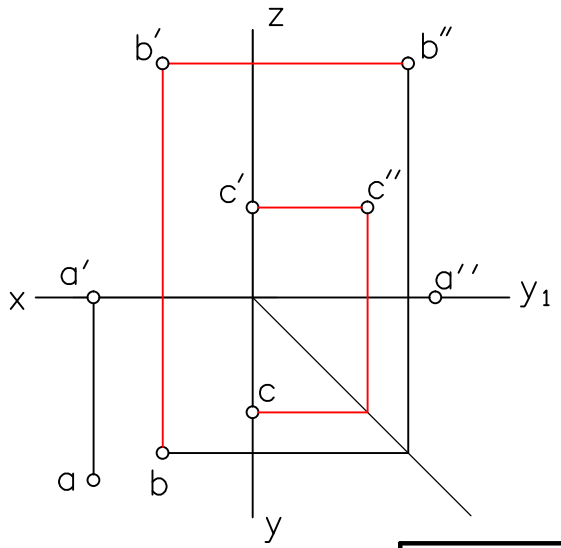


3－7　已知各点的三面投影，试填写其坐标值。

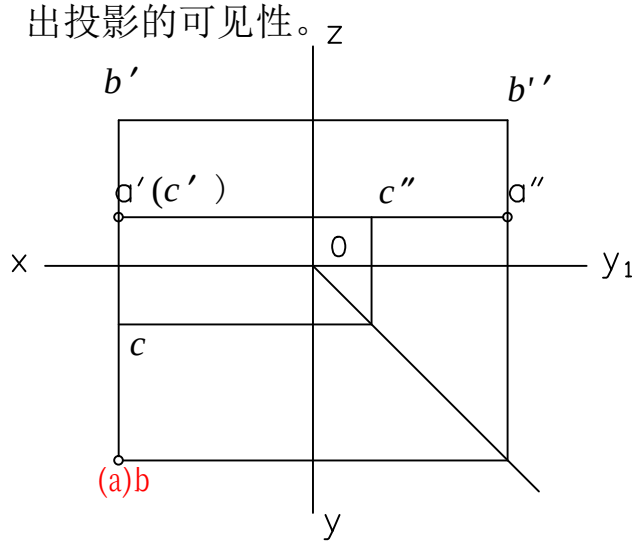


	x	y	z
A	15	15	10
B	20	0	20
C	10	20	0

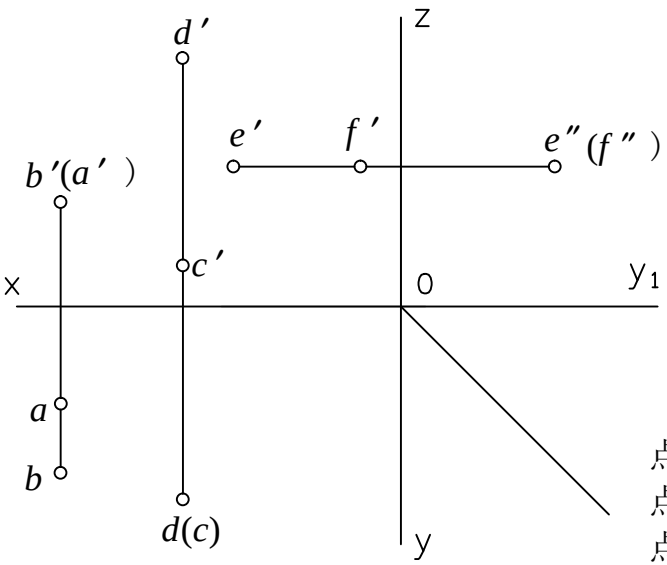
3－8　已知各点的两面投影，求作第三面投影。



3－9　已知A点的三面投影，并知B点在A点的正上方10mm，C点在A点的正后方14mm。求B，C两点的三面投影，并表示出投影的可见性。

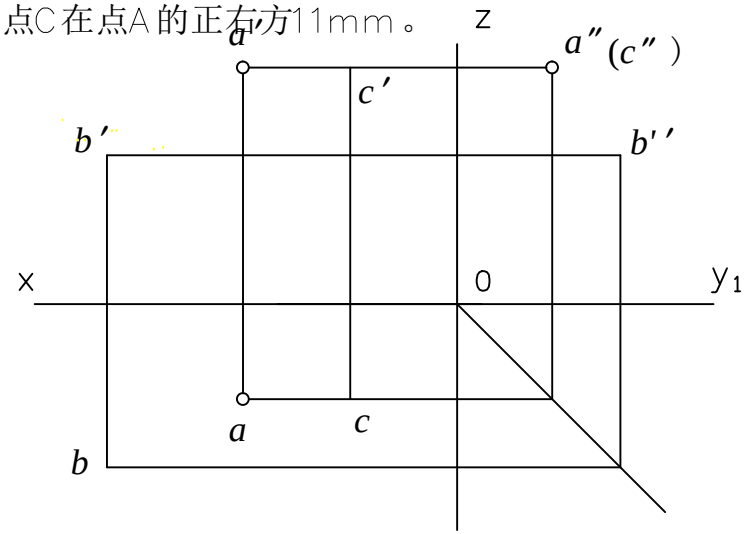


3－10　求各点的第三面投影，并比较其相对位置。

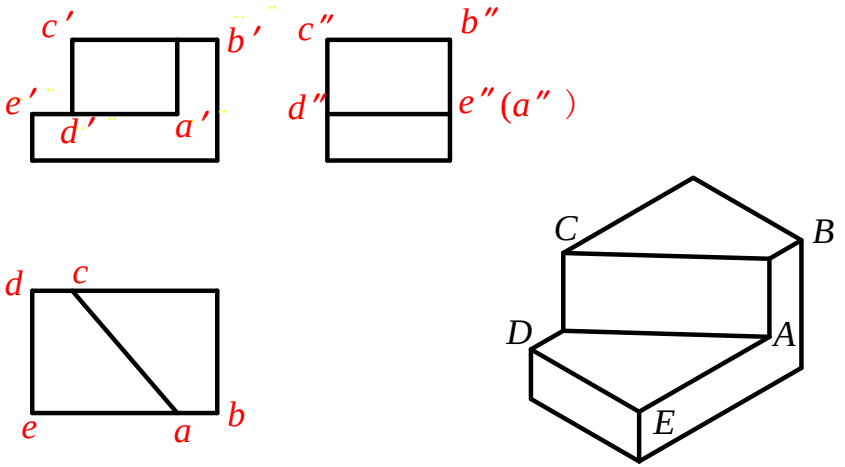


点A在点B正后方6mm
点C在点D正下方20mm
点E在点F正左方13mm

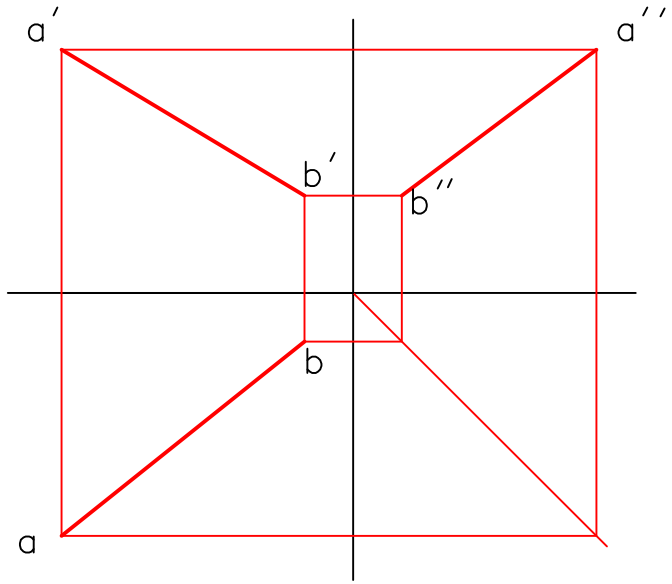
3－11　根据点的相对位置作出两点B、C的投影，并判别重影点的可见性。其中，点B在点A之左14mm、之前7 mm、之下9mm；
点C在点A的正右方11mm。



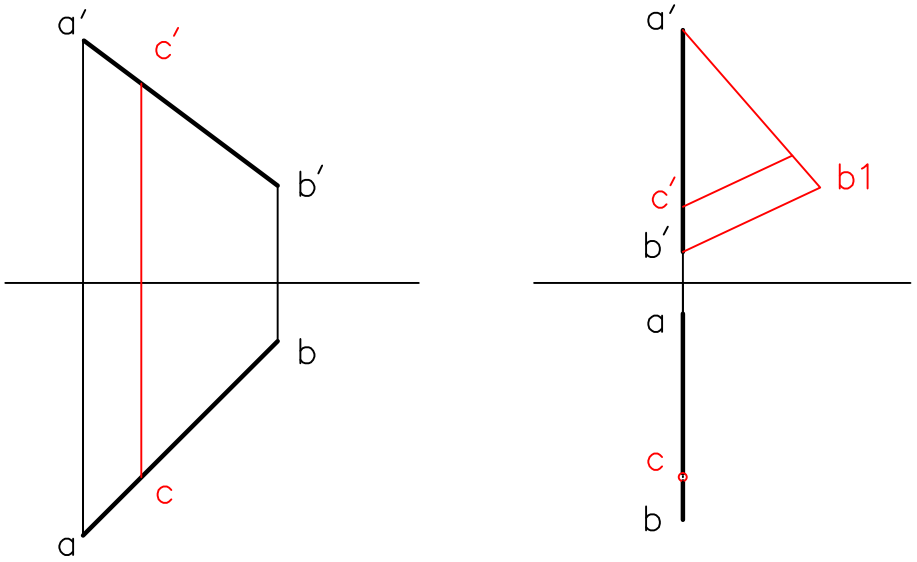
3－12　在物体的投影图中，标出点A、B、C、D、E的投影。



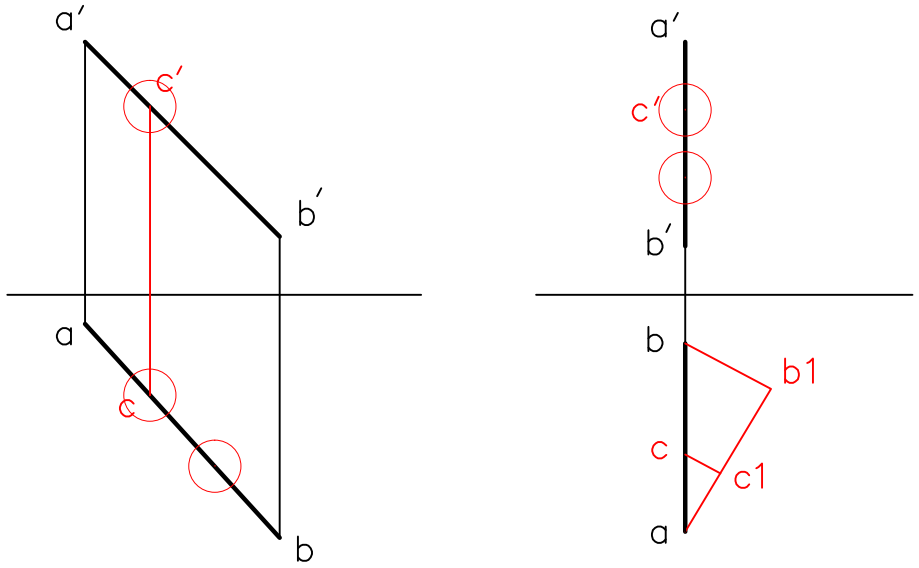
3−13　已知直线AB两端点A(30,25,25)和B(5,5,10)，求作该直线的三面投影图。



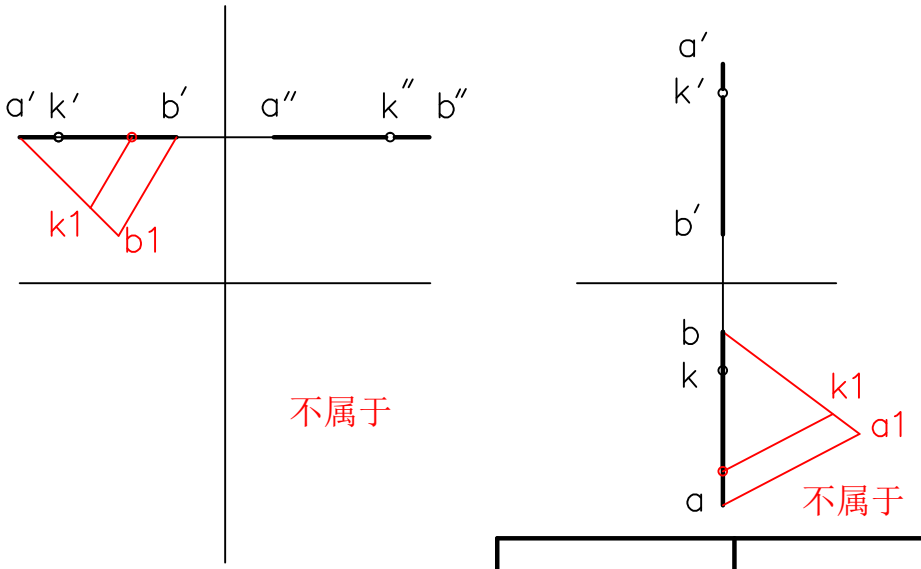
3−14　已知直线AB上一点C距V面20mm求C点的两面投影。



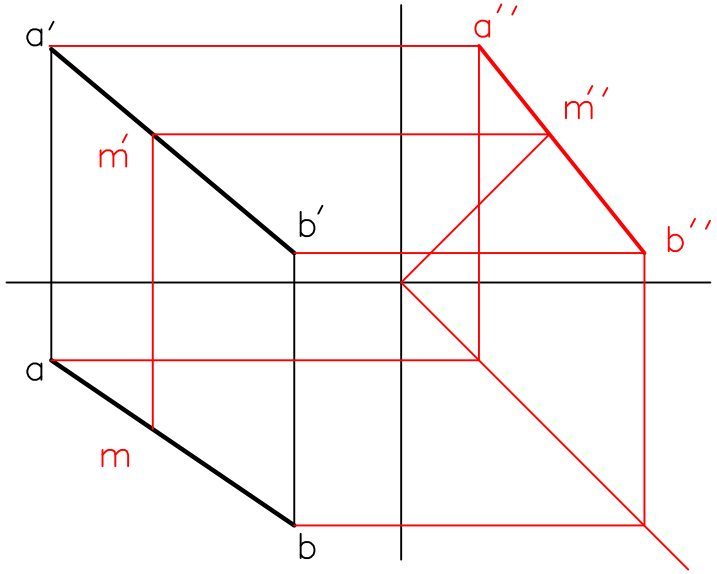
3−15　已知直线AB上一点C，且AC ：CB ＝1 ：2，求C点的两面投影。



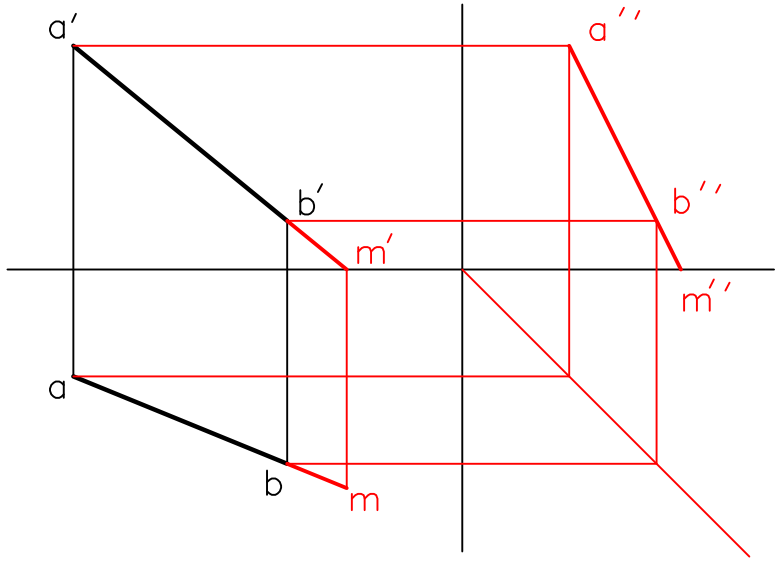
3−16　用作图方法判别K点是否属于直线AB　。



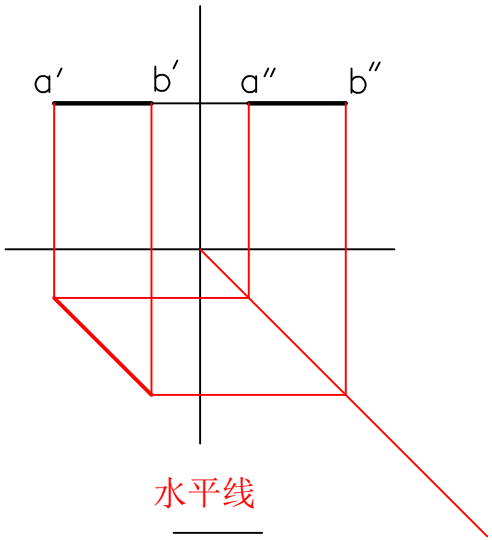
3–17　求直线AB上与V,H面等距离的点M的三面投影。



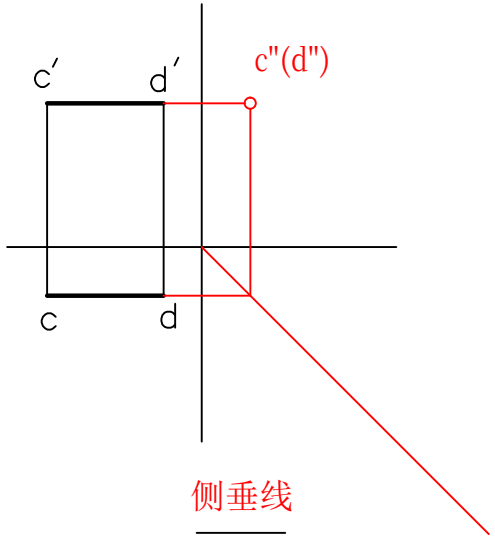
3–18　求位于直线AB上,且又在H面内的M点的三面投影。



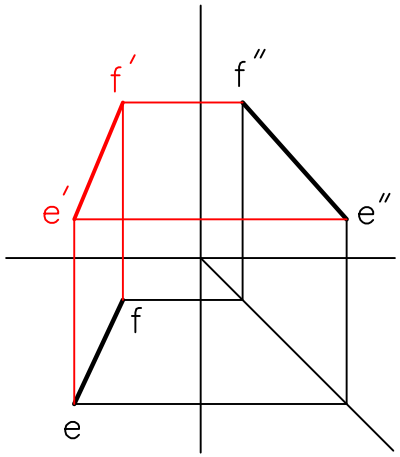
3–19　补画下列各直线的第三面投影,并回答各直线对投影面的相对位置。



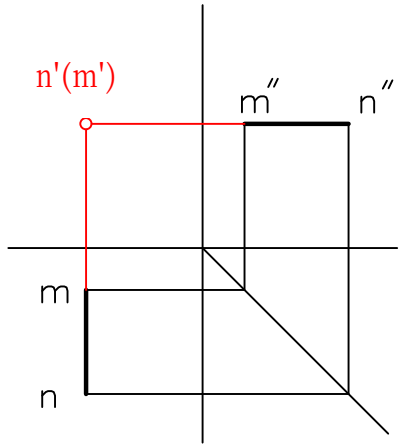
水平线



侧垂线

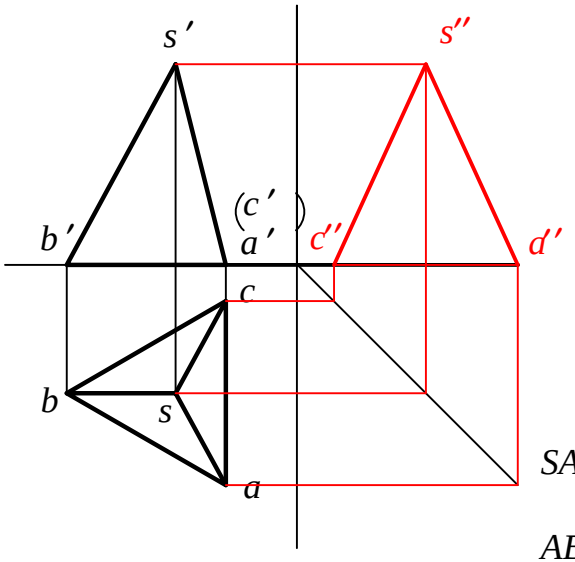


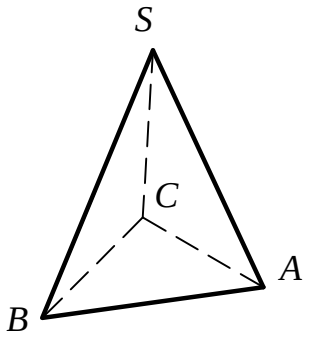
一般直线



正垂线

3－20　判别三棱锥线对投影面的相对位置，并画出第三面投影。

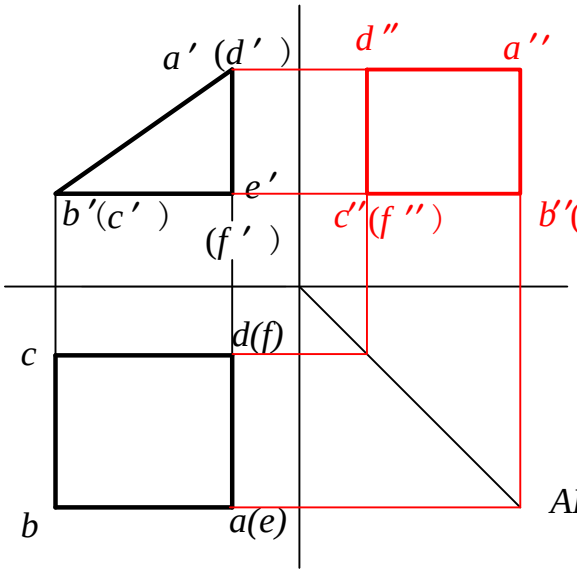


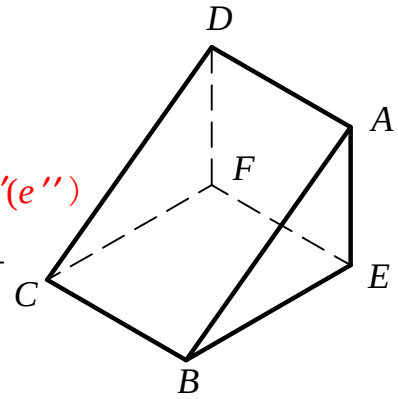


SA是 一般直 线　SB是 正平 线

AB是 水平 线　AC是 正垂 线

3－21　判断三棱柱上直线对投影的位置，并画出第三投影。



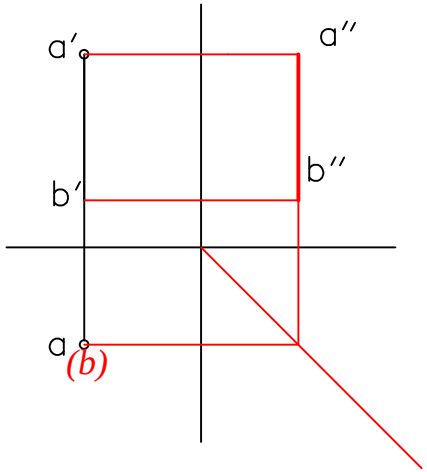


AB是 正平 线　AE是 铅垂 线

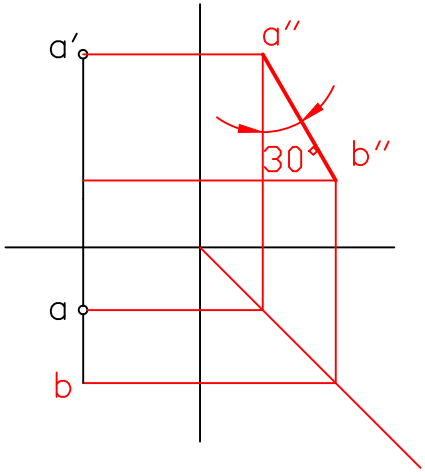
AD是 正垂 线　BE是 侧垂 线

3－22　已知线段AB的长度为15mm，求作它在下列位置时的三面投影。

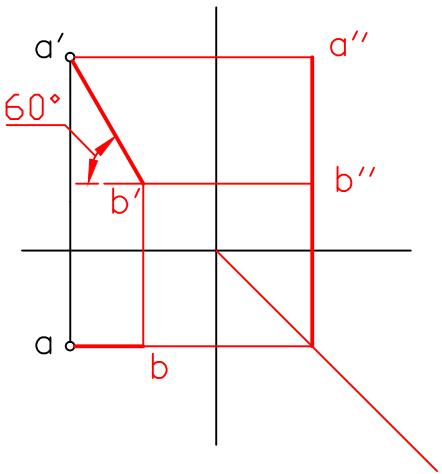
(1) $AB \perp H$



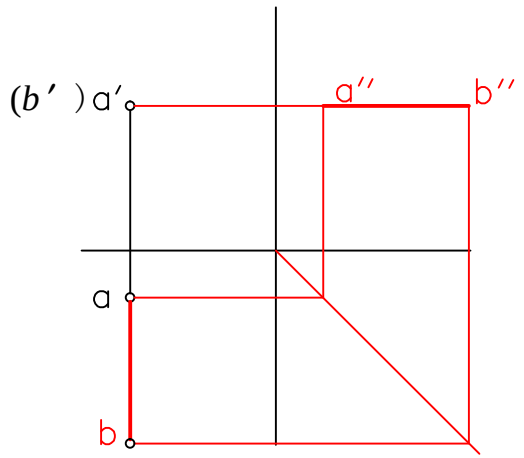
(2) $AB // W, \beta = 30°$



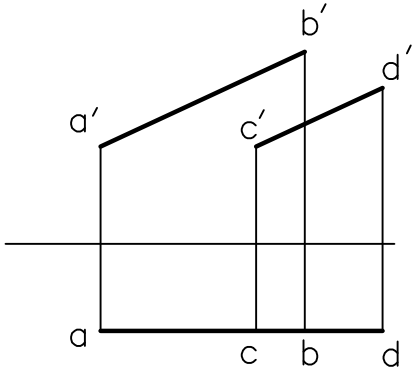
(3) $AB // V, \alpha = 60°$



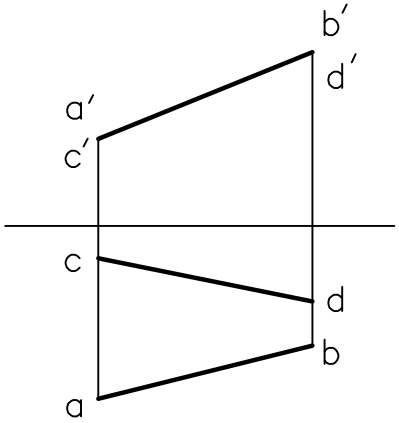
(4) $AB \perp V$



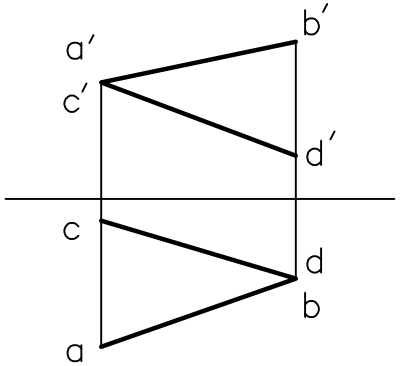
3－23　判别并回答直线AB与CD的空间相对位置。



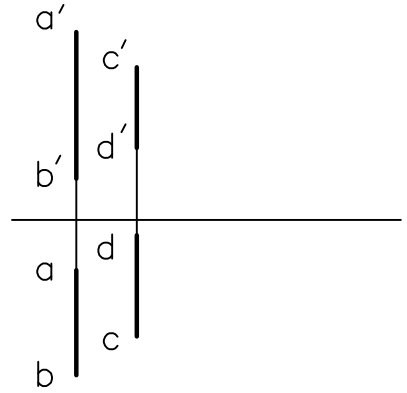
平行



相交

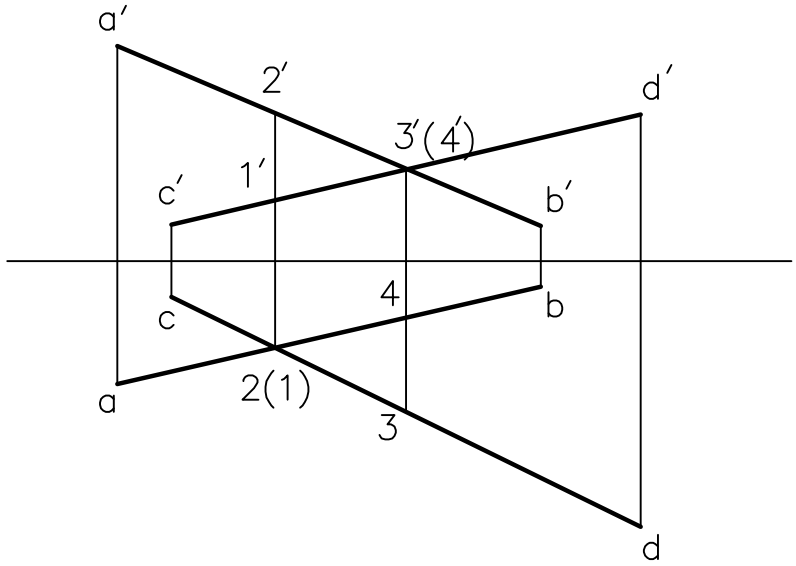


交叉

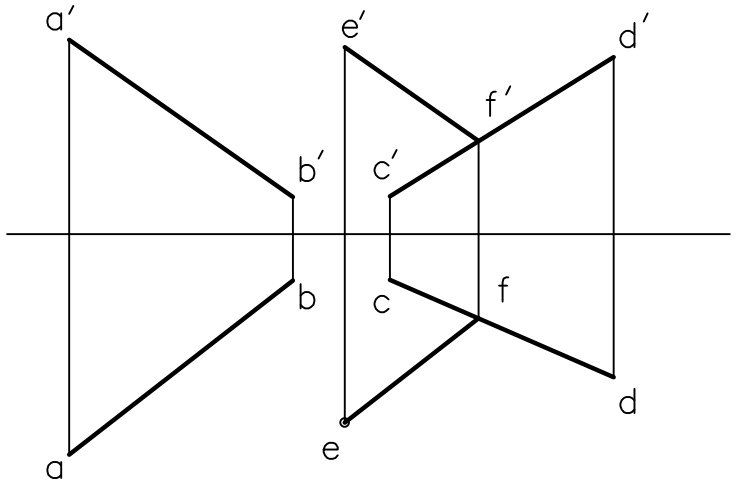


交叉

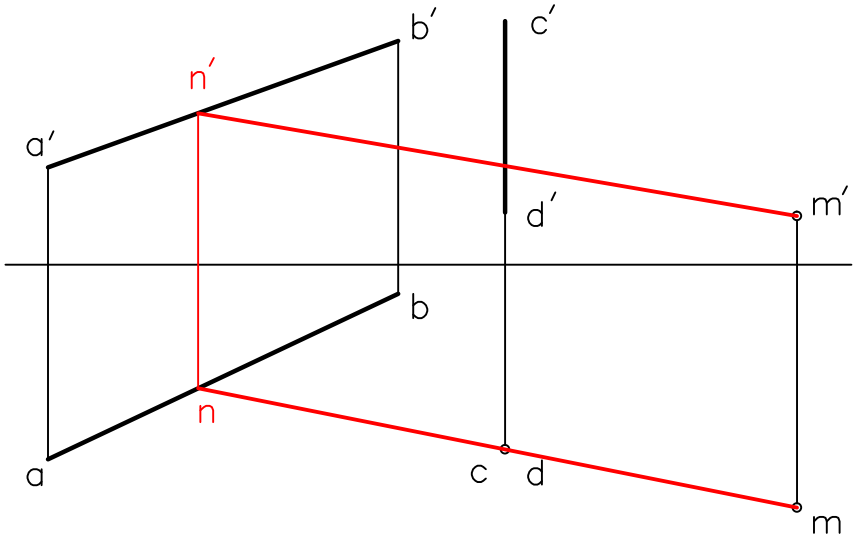
3－24　判别交叉两直线重影点的可见性。



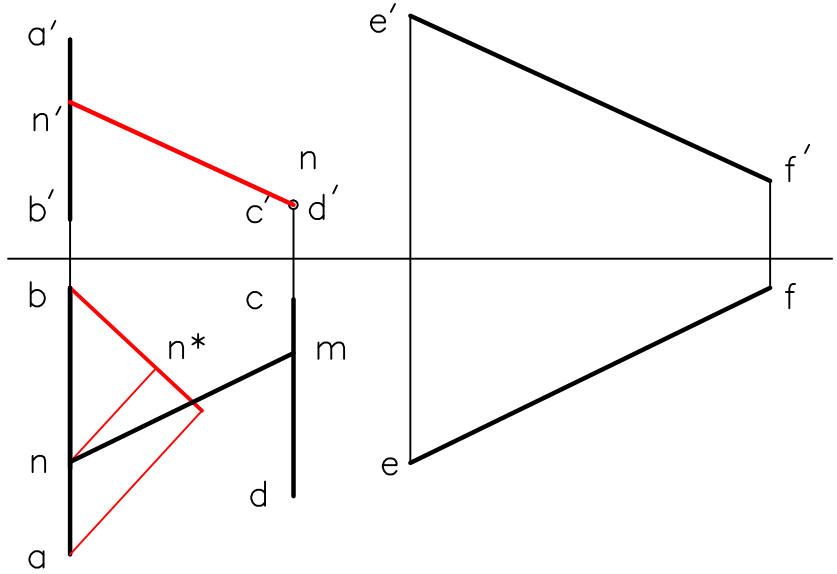
3－25　已知e，试过点E作直线EF，既与AB平行，又与CD相



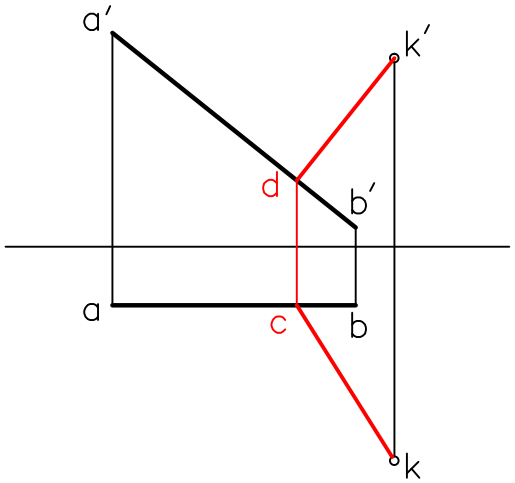
3－26　过点M作直线MN ，令MN与AB ，CD均相交。

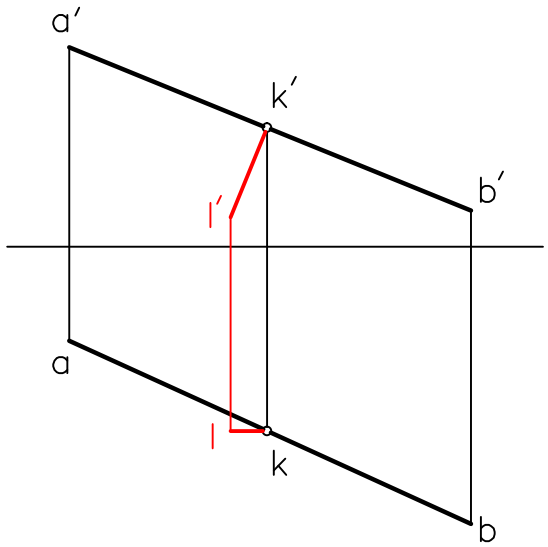


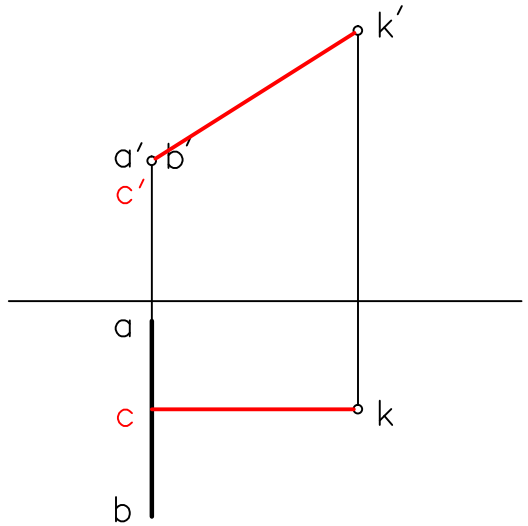
3－27　作一直线MN，令MN平行于EF，并与AB ，CD均相交



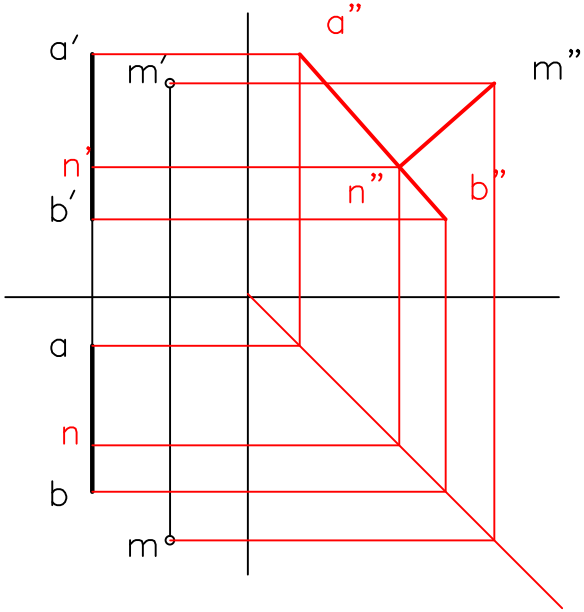
3－28　过点K作一直线与下列直线垂直相交。



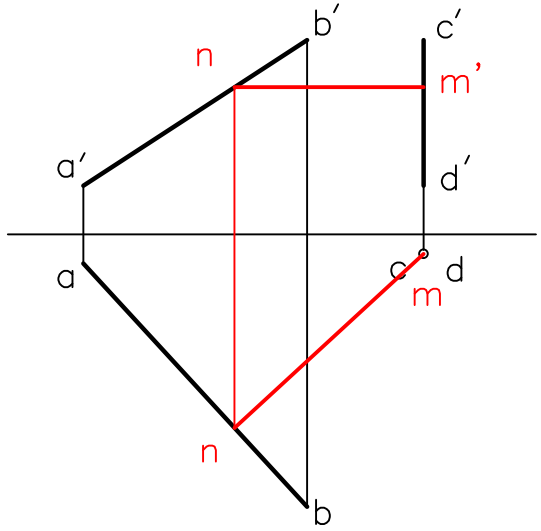




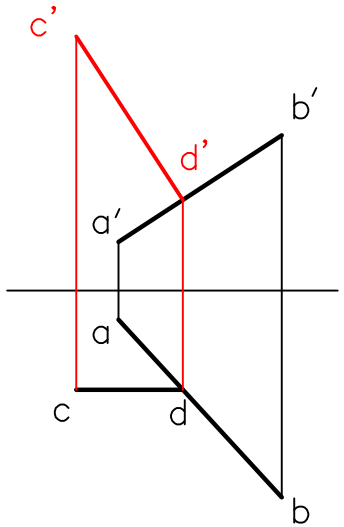
3－29　过M点作一直线与直线AB　垂直相交。



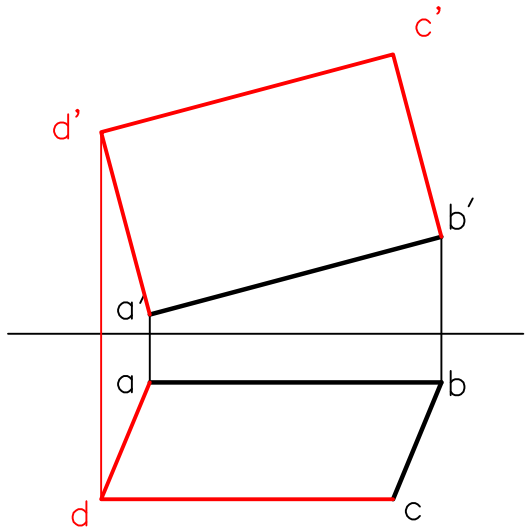
3－30　作一直线MN　，令MN与AB　，CD　垂直相交。



3－31　已知AB　和CD　垂直相交，补画CD的正面投影。

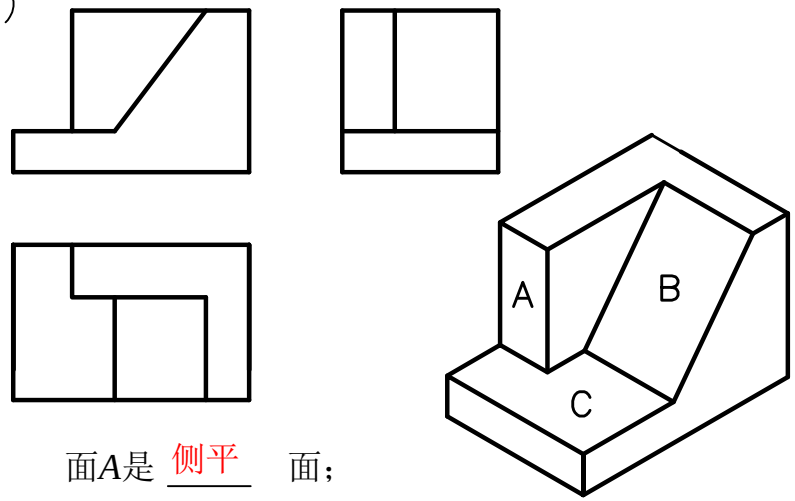


3－32　已知平面ABCD为距形，完成其两面投影。



3－33　在投影图上，标出平面*A*、*B*、*C*的投影，判别并回答各平面对投影面的相对位置。

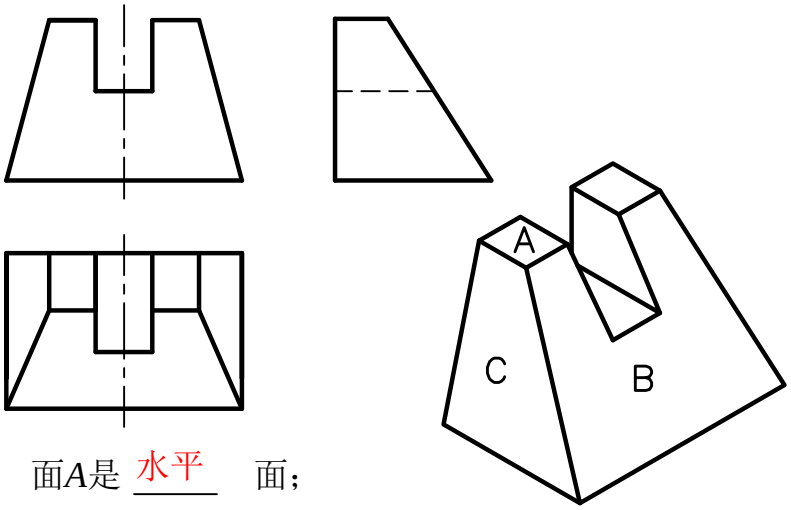
(1)



面*A*是 侧平 面；

面*B*是 正垂 面；　面*C*是 水平 面。

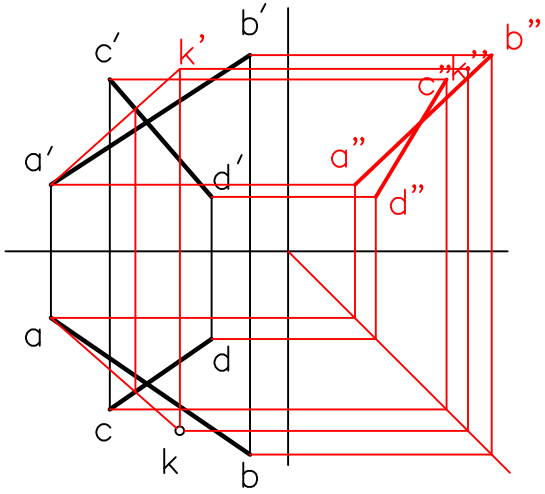
(2)



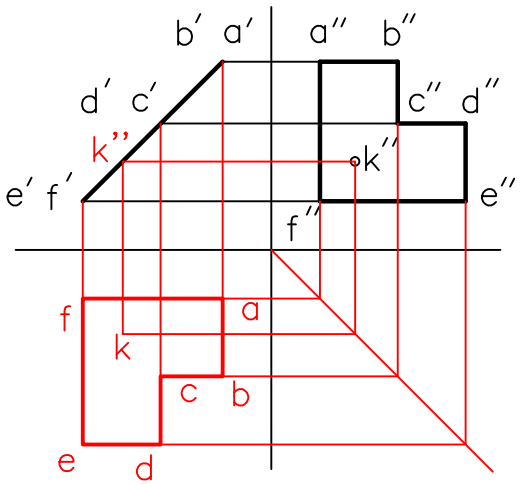
面*A*是 水平 面；

面*B*是 侧垂 面；　面*C*是 正垂 面。

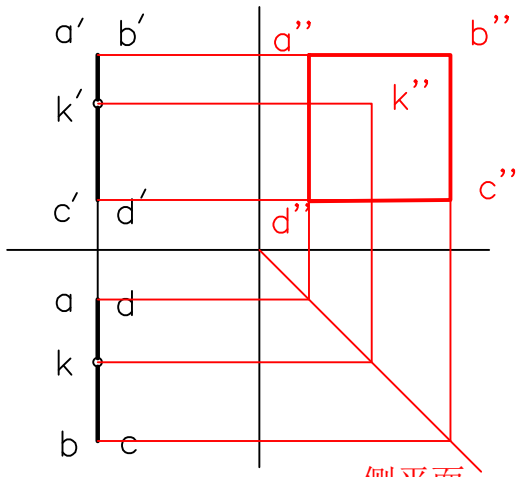
3－34　完成下列平面的第三面投影，求平面内K点的投影，判别并回答各平面对投影面的相对位置。



一般位置



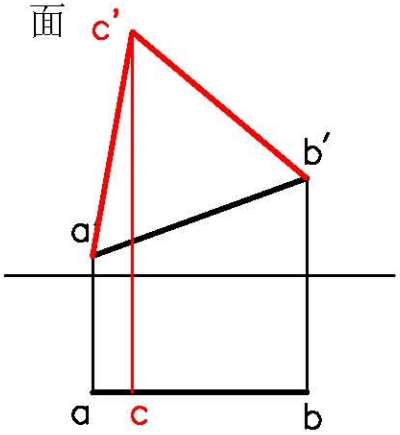
正垂面



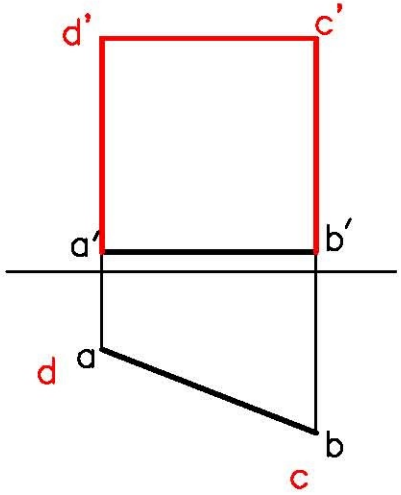
侧平面

3—35 完成下列各平面图形的两面投影。

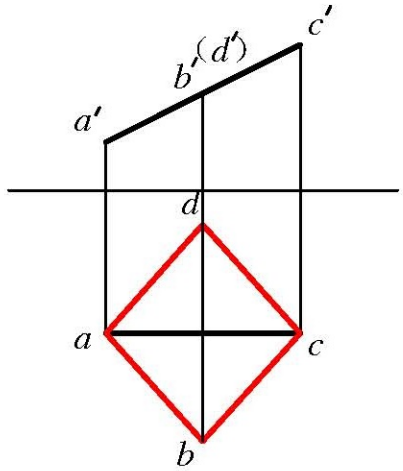
(1)等边△ ABC为正平面



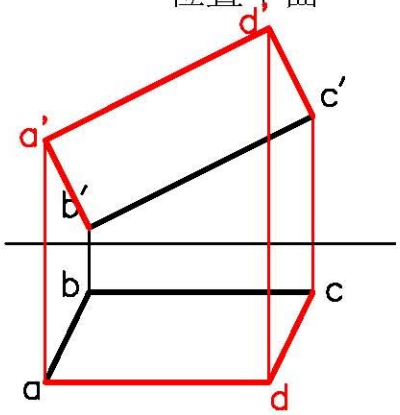
(2)正方形ABCD为铅垂面



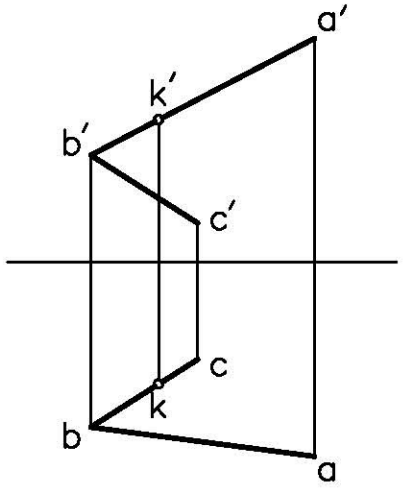
(3)正方形ABCD为正垂面,AC为对角线



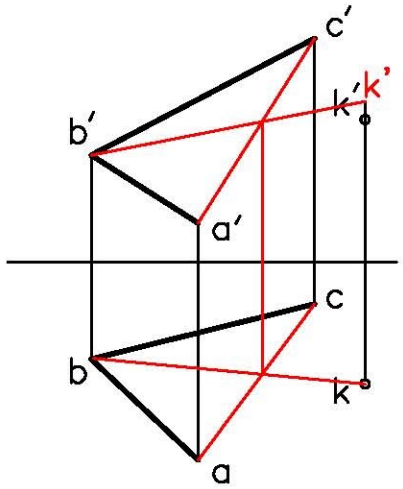
(4)矩形ABCD为一般位置平面



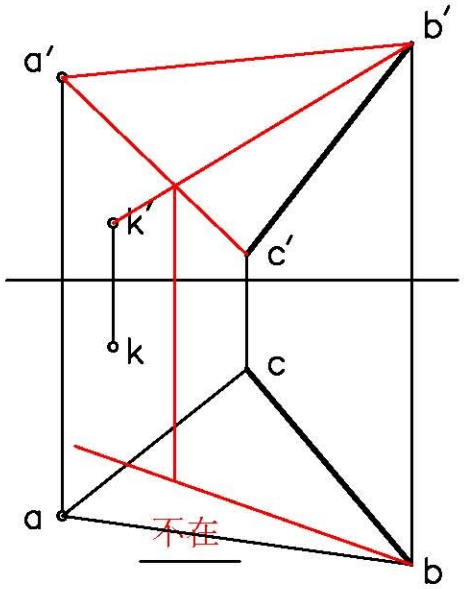
3—36 判别并回答K点是否在下列各平面内。



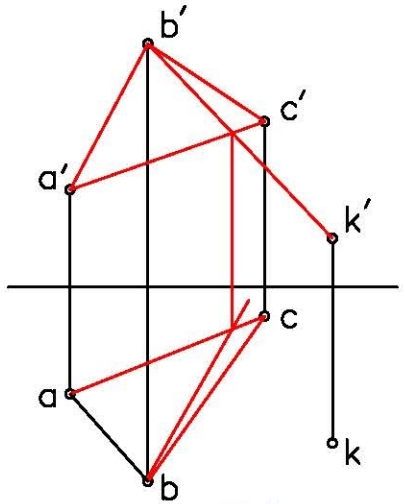
不在



不在

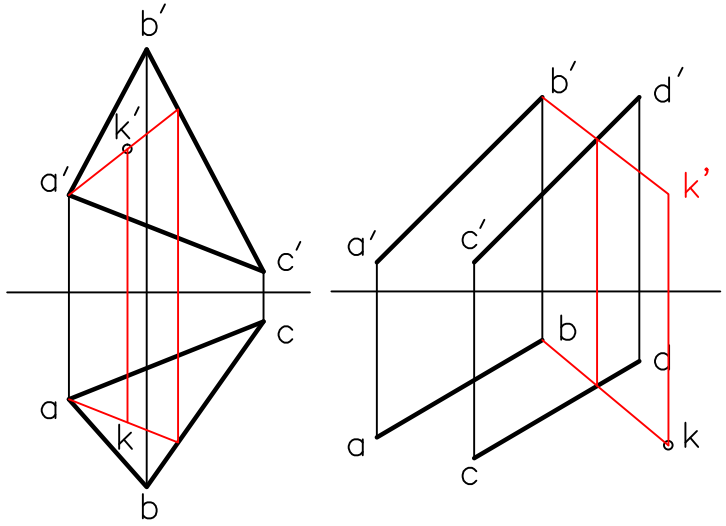


不在

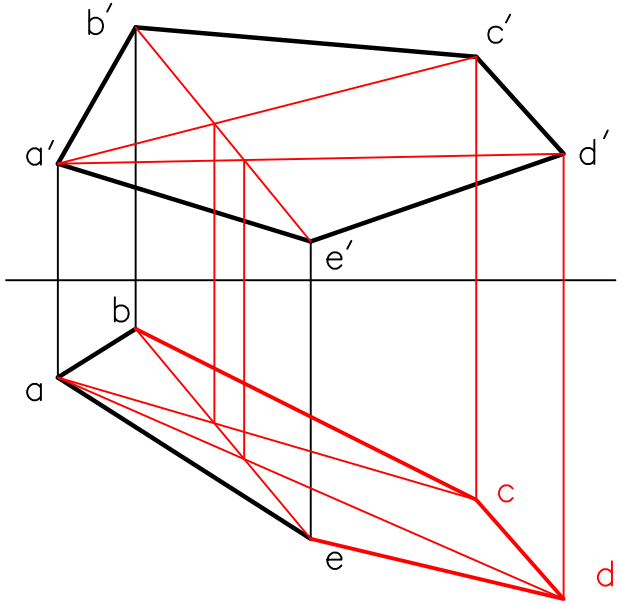


不在

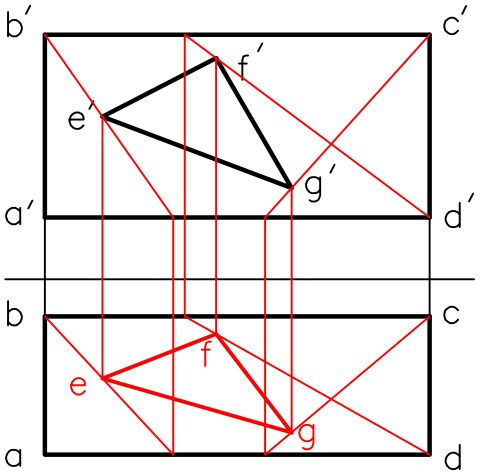
3－37　已知平面内K点的一个投影，求另一个投影。



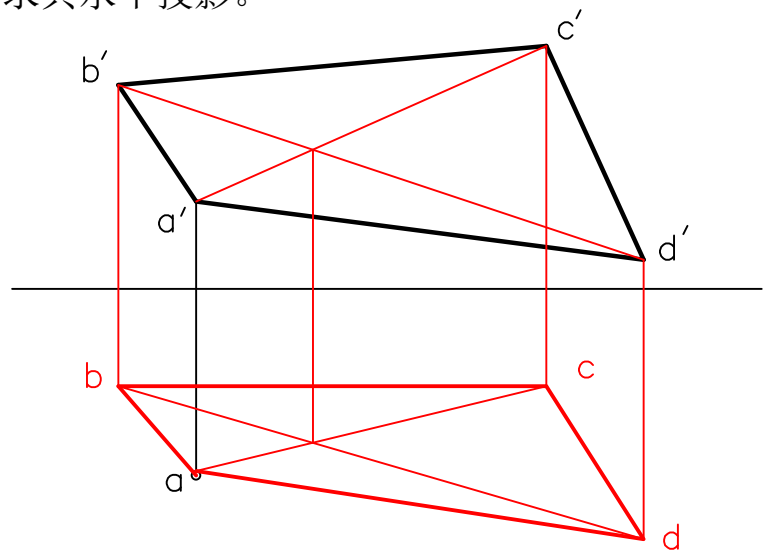
3－38　补全五边平面图形的水平投影。



3－39　求矩形平面ABCD内的△EFG的水平投影。

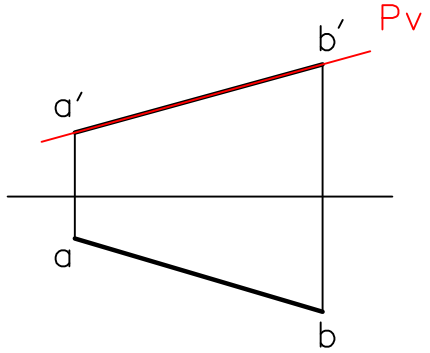


3－40　已知四边形ABCD的BC边为正平线，且距V面为10mm，求其水平投影。

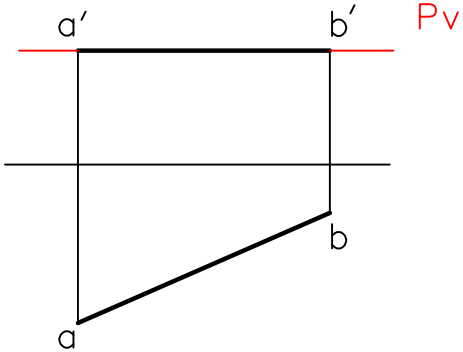


3－41′　过已知直线AB作下列平面（用迹线表示）。

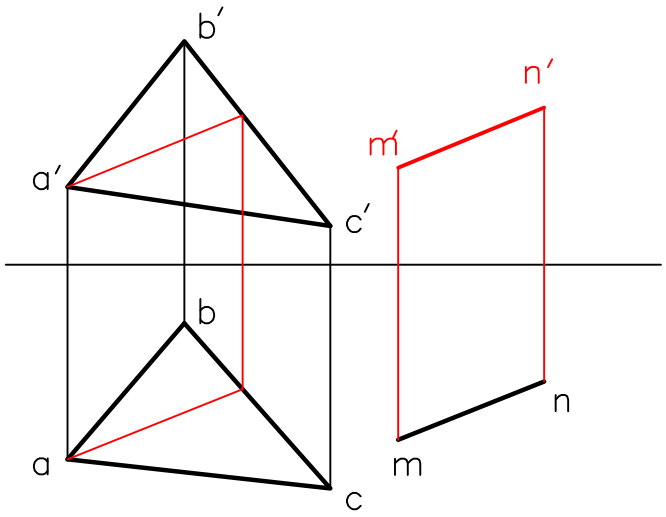
（1）正垂面



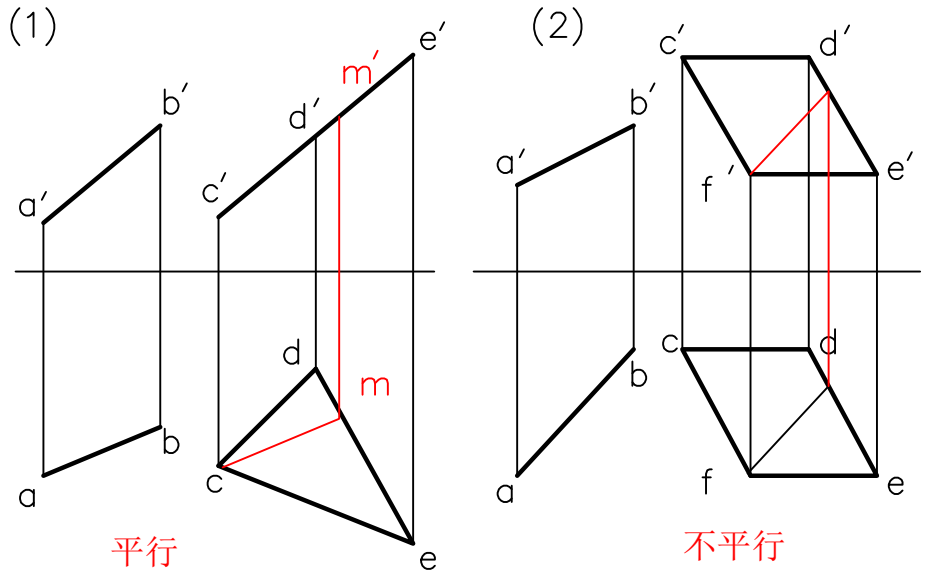
（2）水平面



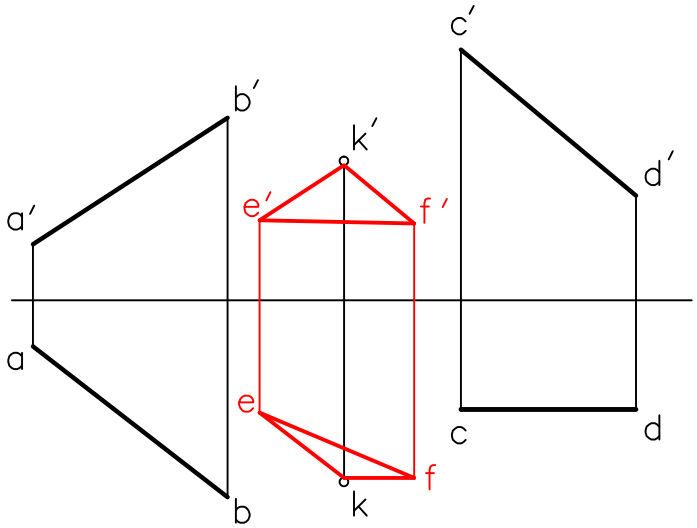
3－42　已知直线MN平行于△ABC，且M点距H面为10mm，求m′ n′。



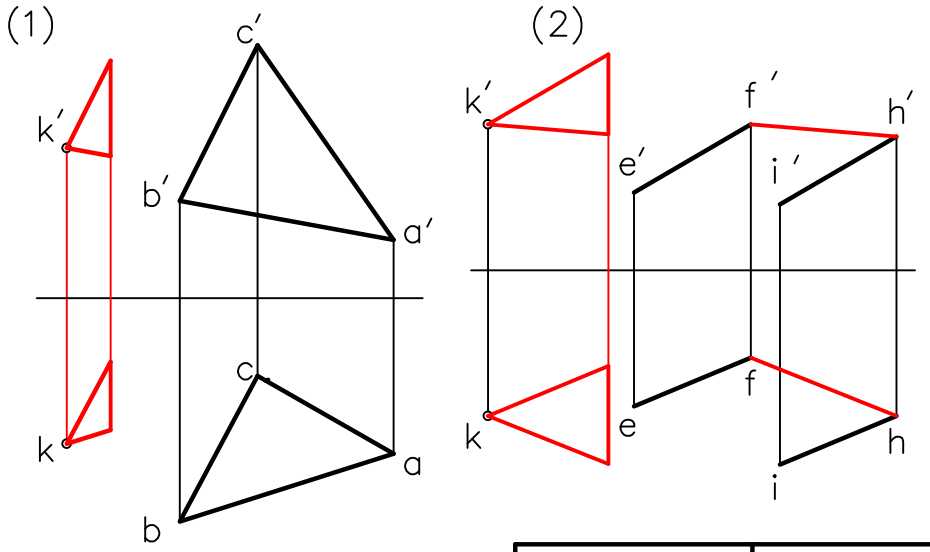
3－43　判别直线AB是否与已知平面平行。



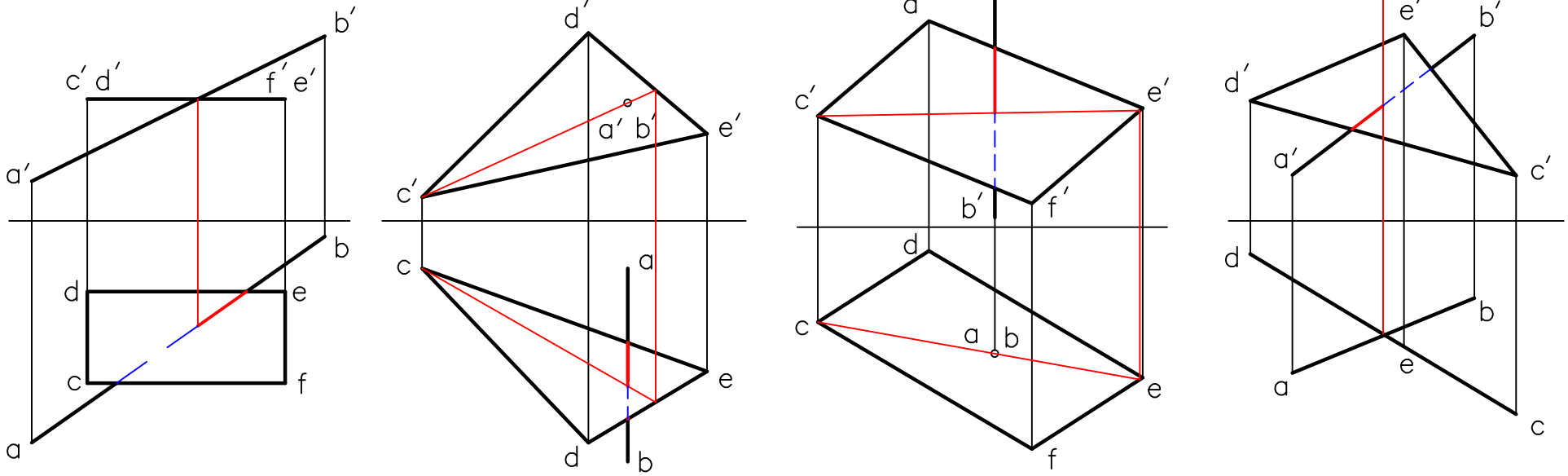
3－44　过K点作三角形平面，平行于两已知直线。



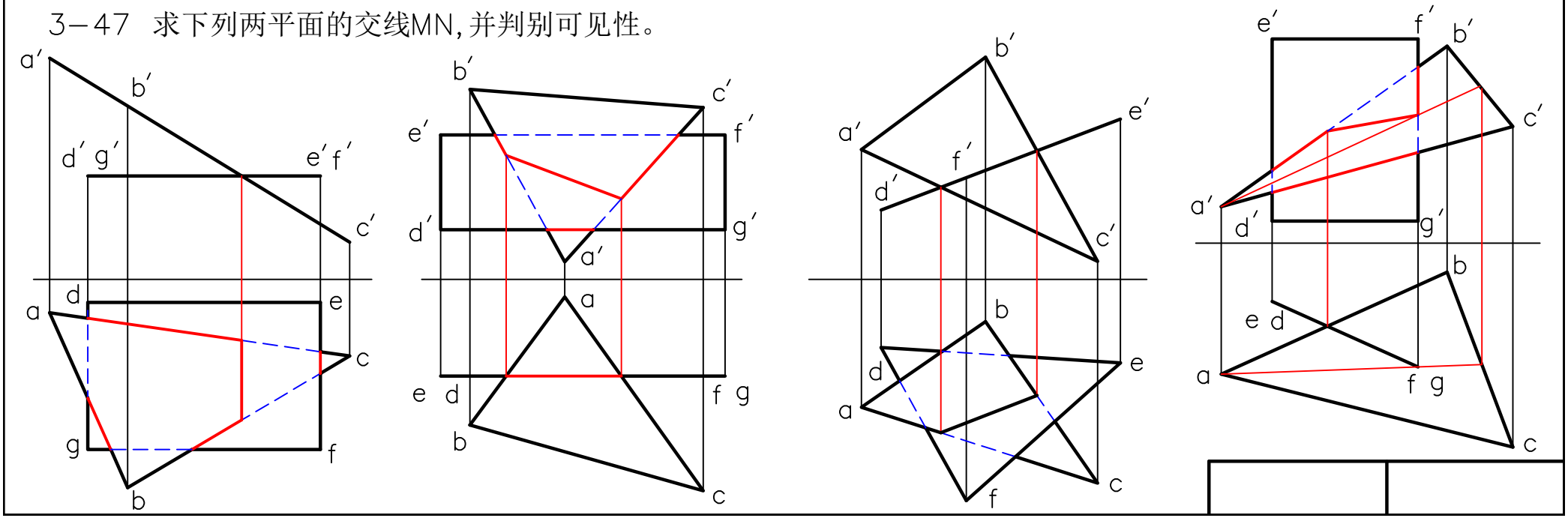
3－45　过K点作平面平行于已知平面。



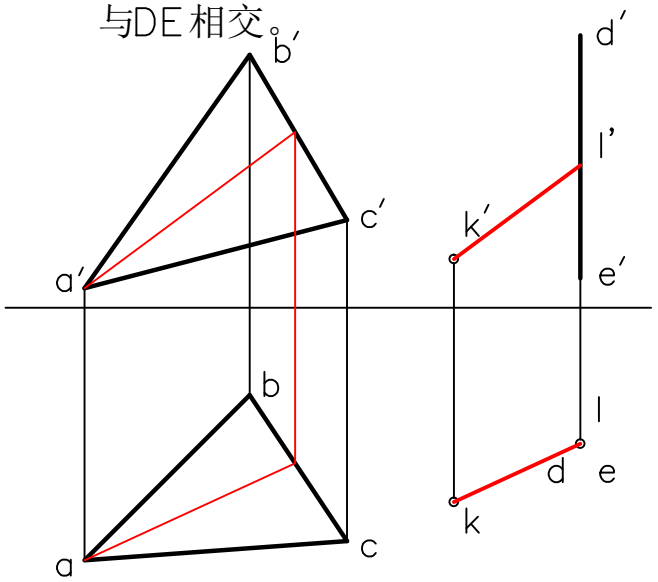
3—46　求下列直线AB与平面的交点K，并判别可见性。



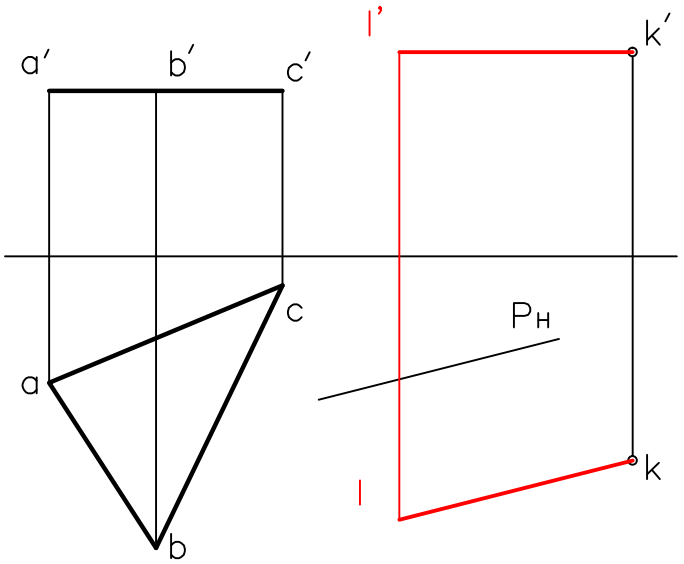
3—47　求下列两平面的交线MN，并判别可见性。



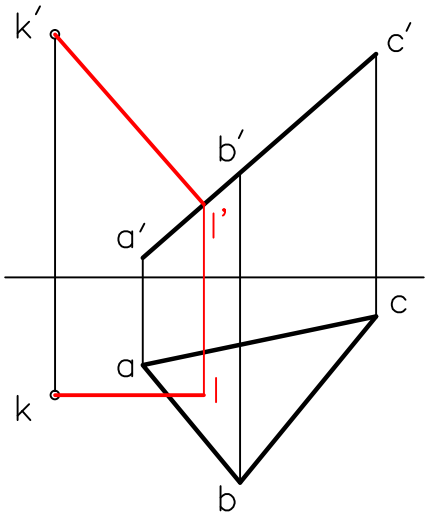
3－48　过K作直线KL，令KL平行于△ABC，并与DE相交。



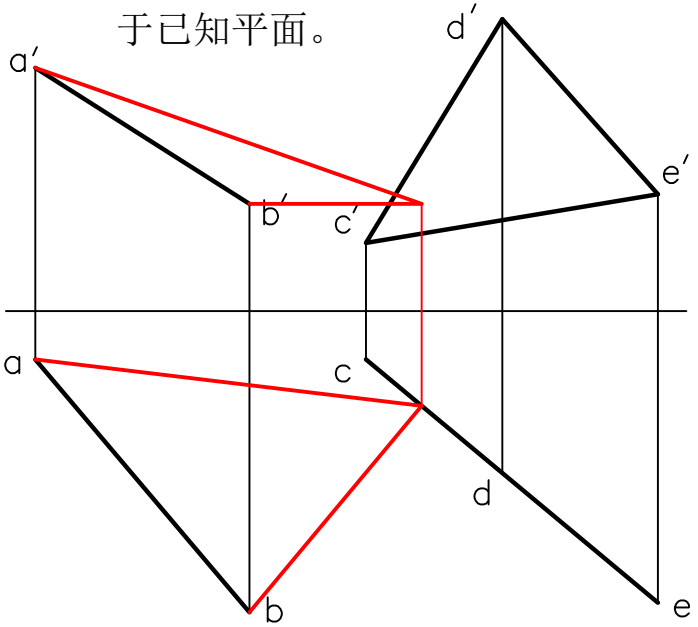
3－49　过K点作直线KL，令KL与△ABC和P面均平行。



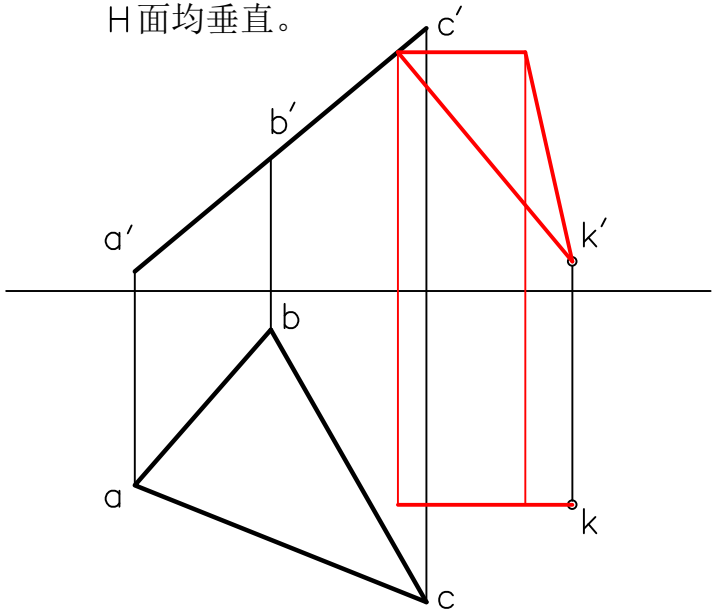
3－50　过K点作平面的垂线KL



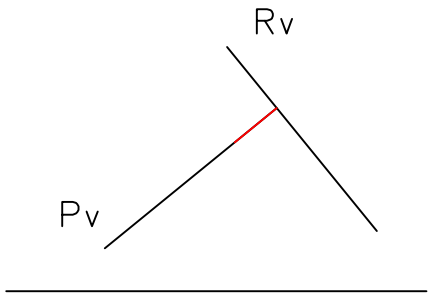
3－51　过直线AB作一三角形平面，使其垂直于已知平面。



3－52　过K点作三角形平面，令其与△ABC及H面均垂直。

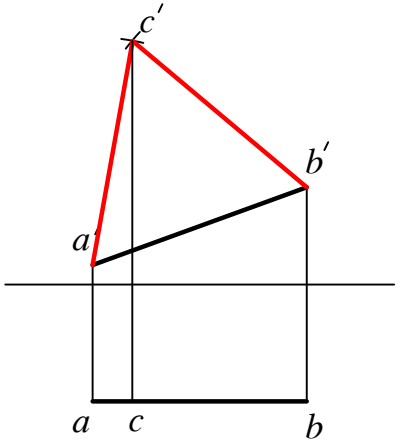


3－53　判别并回答两平面是否垂

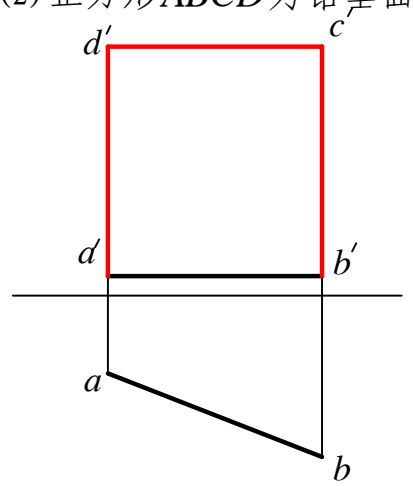


3-35 完成下列各平面图形的两面投影。

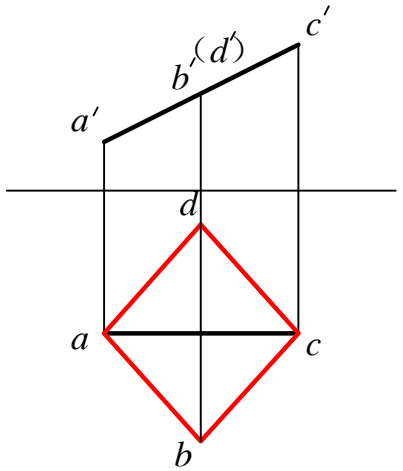
(1) 等边 $\triangle ABC$ 为正平面



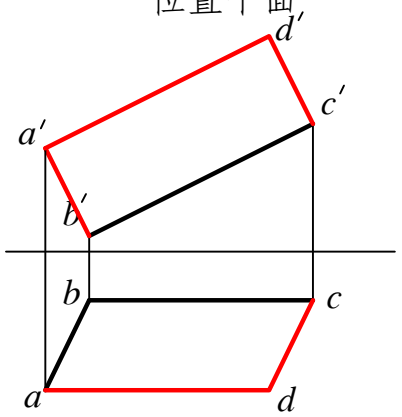
(2) 正方形 $ABCD$ 为铅垂面



(3) 正方形 $ABCD$ 为正垂面，AC 为对角线



(4) 矩形 $ABCD$ 为一般位置平面



3-36 判别并回答 K 点是否在下列各平面内。

